

Mini Project Final Report

ICT1209 – Web Technologies

1. Project Details

PROJECT DETAILS	
Project Title	UniBizHub - University Small Business Advertising Website
Theme	Small Business Advertising Platform for University Students
GitHub Repository	https://github.com/Sigani-glitch/Uni-Biz-Hub
Date	September 07, 2026

Group Members

Members Name	Index Number
S.Sigani	ITT/2024/102
P.T.S.Karunanayaka	ITT/2024/054

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2. Project Overview

2.1 Purpose

Many university students run small businesses like selling snacks, clothing, handmade crafts, or offering services like graphic design. However, they struggle to market their products to others on campus. We created **UniBizHub** to solve this problem by providing a simple website where student sellers can list their items and buyers can find local student services easily.

2.2 Target Audience

- University students running small businesses.
- Students and university staff members who want to buy local products.
- Nearby community members who wish to support student entrepreneurs.

2.3 Key Features

- **User Accounts:** Separate registration and login system for buyers and student sellers.
- **Ad Listing Management:** Sellers can post new business ads, update them, or remove old ones.
- **Categories & Filter:** Browsing products by categories like Food, Clothes, Handmade Goods, Electronics, or Services.
- **Live Search:** Instant keyword searching to find specific items quickly.
- **Direct Seller Contact:** Buyers can directly reach sellers using their phone number, email, or social media links.

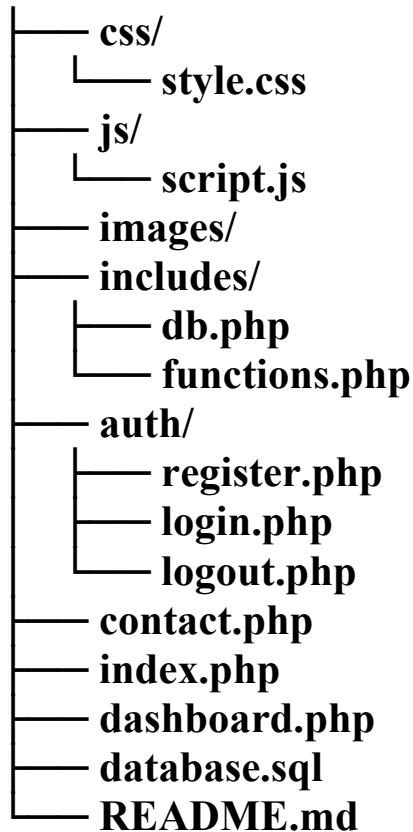
3. System Design

3.1 Database Schema Our project uses a MySQL database with three main tables:

- **users:** Stores id, username, email, hashed password, and role (buyer/seller).
- **advertisements:** Stores id, title, category, price, description, contact_info, and user_id.
- **messages:** Stores id, name, email, message, and created_at from the contact form.

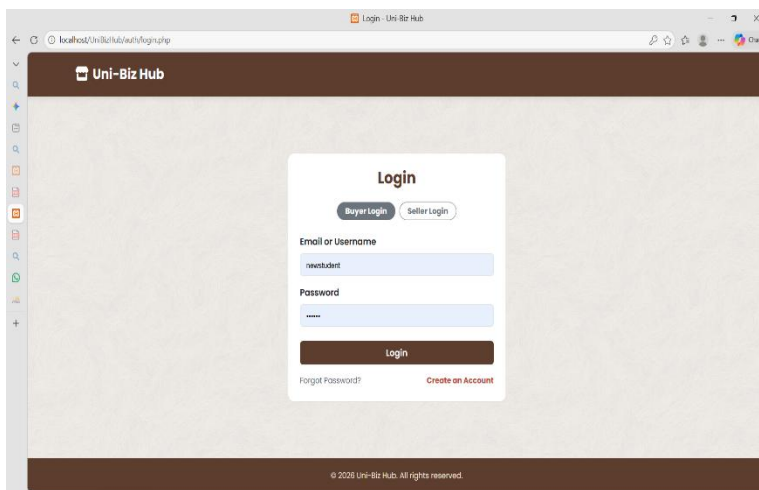
3.2 Directory / Folder Structure

UniBizHub/

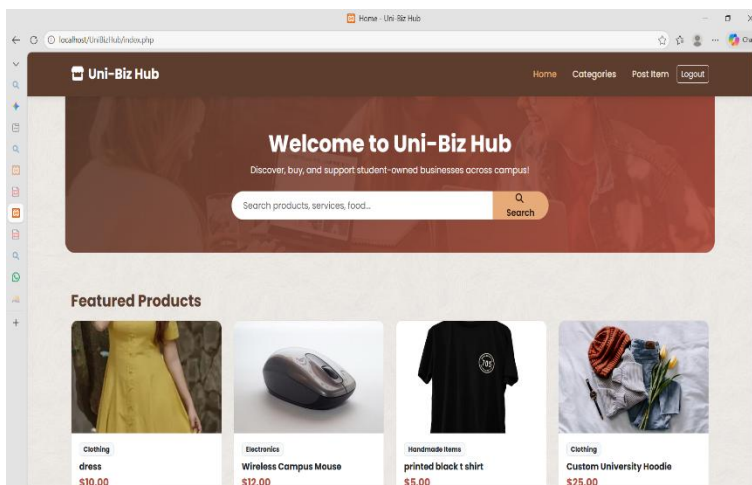


4. Implementation

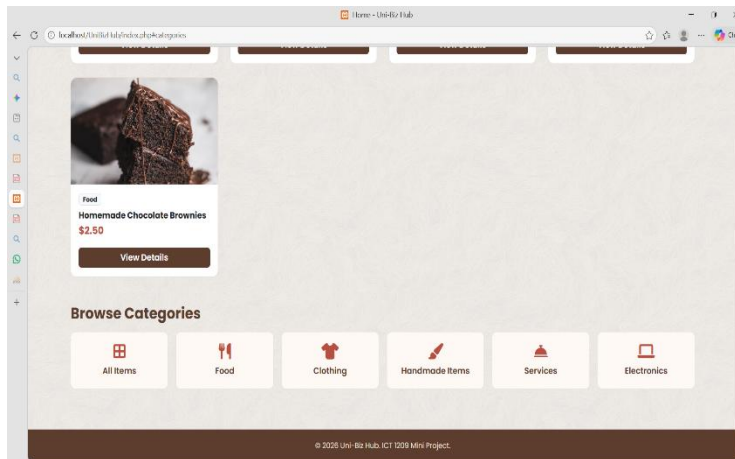
- **User Auth (register & login.php):** We built user registration and login forms. Passwords are encrypted safely using PHP `password_hash()` before storing them in MySQL.
- **Homepage & Search (index.php):** Shows all published ads. We added JavaScript so visitors can filter products by category without refreshing the page.
- **Seller Dashboard (dashboard.php):** Allows logged-in sellers to add new advertisements and manage their active listings.
- **Contact Form (contact.php):** Allows visitors to send queries directly, which are saved into the messages database table.



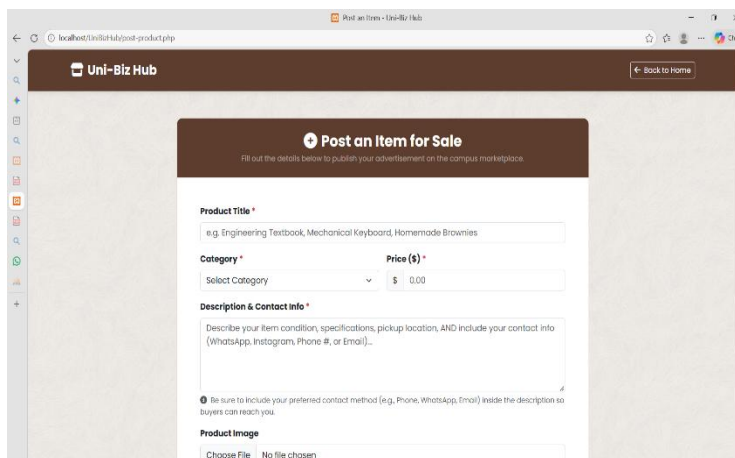
Login Page UI



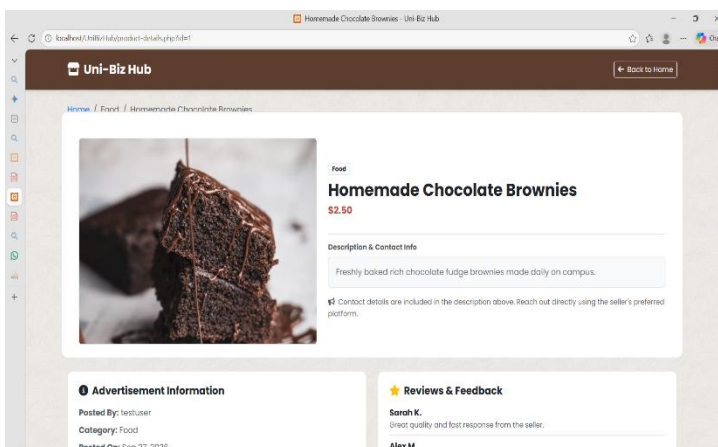
Homepage with Featured Products UI



Homepage with Categories UI



Post an Item Form / Ad Post Form UI



Product Details Page UI

5. Challenges and Solutions

- **Challenge 1: SQL Injection Security**
 - *Problem:* Standard SQL queries are vulnerable to malicious inputs in search and login fields.
 - *Solution:* We implemented PHP PDO Prepared Statements for all database queries to secure input handling.
- **Challenge 2: Mobile Screen Layouts**
 - *Problem:* Web pages looked misaligned on mobile browser screens.
 - *Solution:* We used Bootstrap 5 grid systems (col-md-6, col-sm-12) to make every page fully responsive.
- **Challenge 3: Empty Form Submissions**
 - *Problem:* Users could submit forms without entering required details.
 - *Solution:* We added JavaScript validation on the client side and verified inputs again using PHP on the server side.

6. Individual Contribution

Student Reg No	Key Contributions & Tasks Completed	Contribution %
ITT/2024/054	HTML5 layout structure, Bootstrap styling, JavaScript category filter scripts, client-side form validation, and preparing wireframes & report documentation.	50%
ITT/2024/102	MySQL database creation, PHP authentication logic (login/register/session handling), writing prepared statements, and XAMPP database integration.	50%

7. Setup Instructions

1. Install and open **XAMPP Control Panel**. Click **Start** for both Apache and MySQL modules.
2. Extract or copy the Uni-Biz-Hub project folder into your local directory:
C:/xampp/htdocs/UniBizHub/.
3. Open a web browser and go to <http://localhost/phpmyadmin/>.
4. Create a new database named `unibizhub_db`.
5. Click **Import** and select the `database.sql` file located inside the project folder.
6. Launch the website by visiting <http://localhost/UniBizHub/index.php> in your browser.

8. References

- PHP Official Documentation: <https://www.php.net/manual/en/>
- Bootstrap 5 Framework Documentation: <https://getbootstrap.com/docs/5.0/>
- Mozilla Developer Network (MDN) JavaScript Guide: <https://developer.mozilla.org/en-US/docs/Web/JavaScript>